CS 231. Computer Architecture
Spring 2026
List of Topic questions

1. Binary representation. Byte oriented memory organization. Byte ordering.
2. Encoding Integers. Arithmetic operations with Integers.
3. Encoding Fractional binary numbers (Float, Double). Arithmetic operations with fractional numbers. Rounding.
4. Intel x86 processors. AMD and Intel x64 processors. 32-bit and 64-bit Integer registers.
5. Assembly language. Arithmetic operations in Assembly.
6. Memory addressing modes. MOV, LEA. JMP and conditional jumps. FOR and WHILE loops, SWITCH-CASE in Assembly.
7. Stack organization. Assembly operations with stack. Procedures (functions) in Assembly, transfer of input and output values. Recursive functions. Stack frame.
8. NASM Preprocessor. Single-line and multi-line Macros in NASM.
9. STRUC and ISTRUC. Alignment Principles.
10. Basic data types (db, dw, dd, dq, etc.). Organization of Arrays including Nested Arrays (2D, 3D arrays).
11. Memory layout for running application. Memory read/write transactions.
12. Memory hierarchy. Cache
13. Memory hierarchy. DRAM
14. Memory hierarchy. HDD and SDD
15. Linkers. Types of Symbols. Symbol resolution. Static and dynamic libraries.
16. Asynchronous and Synchronous Exceptions.
17. Processes, Threads. Race conditions. Background jobs.
18. Signals. Signal handlers. Nonlocal jumps.
19. Input/Output. Standard I/O.
20. Virtual Memory. Address translation.
21. Concurrent Programming.
22. Parallelism and Synchronization.
23. Virtual Machines.

Recommended Literature:

1. Lecture Slides
2. Hennessy, J. L., and D. A. Patterson. Computer Architecture: A Quantitative Approach, 7th ed., 2025, ISBN: 9780443154072
3. David A Patterson, John L Hennessy. Computer Organization and Design / The Hardware Software Interface, 5th Edition, 2014, ISBN 0124077269

4. Randal E. Bryant and David R. O'Hallaron, "Computer Systems: A Programmer's Perspective, Second Edition" (CS:APP2e), Prentice Hall, 2011
5. NASM: https://www.nasm.us/pub/nasm/releasebuilds/3.01rc9/doc/nasmdoc.pdf
6. Cache: https://www.youtube.com/watch?v=zF4VMombo7U
   https://www.youtube.com/watch?v=wfVy85Dqiyc
   https://www.youtube.com/watch?v=P_UYI23DddU
7. DRAM: https://www.youtube.com/watch?v=7J7X7aZvMXQ